Modeling with Partial Differential Equations
Exercises - Session 4: Hyperbolic Equations

October 12, 2026

Exercise 1 — Linear transport and Hamiltonian systems Let B be a real 2×2 matrix. For $x = (x_1, x_2) \in \mathbb{R}^2$, set $b(x) = Bx = (b_1(x), b_2(x))$. We consider the transport equation

$$\begin{cases} \partial_t u(t,x) + b_1(x)\partial_{x_1} u(t,x) + b_2(x)\partial_{x_2} u(t,x) = 0, & t > 0, \quad x \in \mathbb{R}^2, \\ u(0,x) = u_0(x), & x \in \mathbb{R}^2, \end{cases} \tag{1}$$

where $u_0 \in C^1(\mathbb{R}^2)$. Recall that the characteristic curves satisfy $\dot{X}(t) = BX(t)$, and that the solution of this linear system with $X(0) = y$ is $X(t) = e^{tB}y$.

1. **Conservation of mass and quadratic energy**
   Assume that $\operatorname{div} b = 0$, where $\operatorname{div} b = \partial_{x_1} b_1 + \partial_{x_2} b_2$. Let $\Phi : \mathbb{R} \to \mathbb{R}$ be a continuous function such that

   $$\int_{\mathbb{R}^2} |\Phi(u_0(x))| \, \mathrm{d}x < +\infty.$$

   Show that, for every $t \geq 0$,

   $$\int_{\mathbb{R}^2} |\Phi(u(t,x))| \, \mathrm{d}x < +\infty, \qquad \int_{\mathbb{R}^2} \Phi(u(t,x)) \, \mathrm{d}x = \int_{\mathbb{R}^2} \Phi(u_0(x)) \, \mathrm{d}x.$$

   Hint (change of variables). *Let $F : \mathbb{R}^2 \to \mathbb{R}^2$ be a continuously differentiable bijection whose inverse is also continuously differentiable. For a continuous function g, the change of variables $x = F(y)$ gives*

   $$\int_{\mathbb{R}^2} g(x) \, \mathrm{d}x = \int_{\mathbb{R}^2} g(F(y)) \, |\det DF(y)| \, \mathrm{d}y,$$

   whenever either integral is absolutely convergent; in that case, the other is absolutely convergent as well. Here, for $y = (y_1, y_2)$,

   $$DF(y) = \begin{pmatrix} \partial_{y_1} F_1(y) & \partial_{y_2} F_1(y) \\ \partial_{y_1} F_2(y) & \partial_{y_2} F_2(y) \end{pmatrix}.$$

   The same formula applies to $|g|$.

   We also recall that, for any square matrix M, $\det(e^M) = e^{\operatorname{tr} M}$.[1] Deduce that the total mass $M(t) = \int_{\mathbb{R}^2} u(t,x) \, \mathrm{d}x$ is constant whenever $\int_{\mathbb{R}^2} |u_0(x)| \, \mathrm{d}x$ is finite, and that the quadratic energy

   $$E_2(t) = \frac{1}{2} \int_{\mathbb{R}^2} |u(t,x)|^2 \, \mathrm{d}x$$

   remains finite and constant whenever $E_2(0) < +\infty$.

   Solution : *Set $X(t) = e^{tB}y$. Along this characteristic, $\dot{X}_1(t) = b_1(X(t))$ and $\dot{X}_2(t) = b_2(X(t))$. The chain rule and the equation give*

   $$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t} u(t, X(t)) &= \partial_t u(t, X(t)) + \partial_{x_1} u(t, X(t)) \dot{X}_1(t) + \partial_{x_2} u(t, X(t)) \dot{X}_2(t) \\ &= 0. \end{aligned}$$

[1] For an upper triangular matrix T, the matrix e^T is upper triangular with diagonal entries $e^{T_{jj}}$, so $\det(e^T) = e^{\operatorname{tr} T}$. Every complex square matrix is similar to an upper triangular matrix: $M = PTP^{-1}$ with P invertible. Since $e^M = Pe^T P^{-1}$ and trace and determinant are unchanged by similarity, the identity follows.

Thus $u(t, e^{tB}y) = u_0(y)$, and since e^{tB} is invertible,

$$u(t, x) = u_0(e^{-tB}x).$$

Conversely, this formula defines a C^1 solution, so it also proves uniqueness among classical solutions.

Since $b_1(x) = B_{11}x_1 + B_{12}x_2$ and $b_2(x) = B_{21}x_1 + B_{22}x_2$, we have $\operatorname{div} b = B_{11} + B_{22} = \operatorname{tr} B$. The determinant identity for a matrix exponential yields

$$\det(e^{tB}) = e^{t \operatorname{tr} B} = 1.$$

The change of variables $x = e^{tB}y$ therefore gives

$$\int_{\mathbb{R}^2} |\Phi(u(t, x))| \, \mathrm{d}x = \int_{\mathbb{R}^2} |\Phi(u_0(y))| \, \mathrm{d}y < +\infty.$$

Applying the same change of variables without the absolute value proves the claimed conservation law. In particular, taking $\Phi(s) = s$ and $\Phi(s) = s^2/2$ gives $M(t) = M(0)$ and $E_2(t) = E_2(0)$ under the respective assumptions.

2. **Hamiltonian systems**
Write $x = (q, p) \in \mathbb{R}^2$, where q and p are real numbers. Given a function $H \in C^2(\mathbb{R}^2)$, the associated *Hamiltonian system* is

$$\dot{q} = \partial_p H(q, p), \qquad \dot{p} = -\partial_q H(q, p).$$

Show that $H(q(t), p(t))$ is constant along every solution, on its interval of existence. We now restrict to

$$H(x) = \frac{1}{2} x^\top S x, \qquad S \in \mathbb{R}^{2\times 2}, \quad S^\top = S.$$

With $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, write the Hamiltonian velocity $b(q, p) = (\partial_p H(q, p), -\partial_q H(q, p))$ as $b(x) = Bx$, determine B, and verify that $\operatorname{div} b = 0$. Deduce the conservation laws of Question 1 for the transport equation associated with this system.

Solution : *The two equations give*

$$\frac{\mathrm{d}}{\mathrm{d}t} H(q(t), p(t)) = \partial_q H \, \dot{q} + \partial_p H \, \dot{p} = \partial_q H \, \partial_p H - \partial_p H \, \partial_q H = 0.$$

Write the symmetric matrix as

$$S = \begin{pmatrix} \alpha & \beta \\ \beta & \gamma \end{pmatrix}.$$

Then $H(q, p) = \frac{1}{2}(\alpha q^2 + 2\beta qp + \gamma p^2)$, so

$$\partial_q H = \alpha q + \beta p, \qquad \partial_p H = \beta q + \gamma p.$$

Consequently $b(q, p) = (\beta q + \gamma p, -\alpha q - \beta p)$ and

$$B = \begin{pmatrix} \beta & \gamma \\ -\alpha & -\beta \end{pmatrix} = JS, \qquad \operatorname{div} b = \partial_q(\beta q + \gamma p) + \partial_p(-\alpha q - \beta p) = \beta - \beta = 0.$$

Therefore Question 1 applies to this velocity. Every integral of the form $\int_{\mathbb{R}^2} \Phi(u(t, x)) \, \mathrm{d}x$ is conserved whenever $\int_{\mathbb{R}^2} |\Phi(u_0(x))| \, \mathrm{d}x$ is finite; in particular, this holds for the mass and quadratic energy under their stated assumptions. Moreover, $H(e^{tB}x) = H(x)$ along the globally defined linear flow.

Exercise 2 — Normal modes of a string with fixed endpoints Let $L > 0$ and $c > 0$. We consider the homogeneous wave equation for a string whose endpoints are fixed:

$$\begin{cases} \partial_{t,t} u - c^2 \partial_{x,x} u = 0, & t > 0, \quad 0 < x < L, \\ u(t, 0) = u(t, L) = 0, & t \geq 0, \\ u(0, x) = u_0(x), \quad \partial_t u(0, x) = v_0(x), & 0 \leq x \leq L. \end{cases} \tag{2}$$

We first construct separated solutions, then use superposition to match the initial data.

1. **Separation of variables**
Look for a nonzero separated solution $u(t,x) = A(t)B(x)$. Show that there is a constant $\lambda \in \mathbb{R}$ such that

$$B'' + \lambda B = 0, \qquad B(0) = B(L) = 0, \qquad A'' + c^2 \lambda A = 0.$$

Solution : *Substitution gives $A''(t)B(x) = c^2 A(t)B''(x)$. Choose x_* such that $B(x_*) \neq 0$ and define $\lambda = -B''(x_*)/B(x_*)$. Evaluating the identity at x_* gives $A'' + c^2\lambda A = 0$ for every t. Next choose t_* such that $A(t_*) \neq 0$; the same identity now gives $B'' + \lambda B = 0$ for every x. Evaluating the boundary conditions at t_* yields $B(0) = B(L) = 0$. This argument remains valid at the zeros of either factor.*

2. **Spatial profiles**
Show that the spatial problem has no nonzero solution when $\lambda \leq 0$. For $\lambda > 0$, determine all admissible values and obtain, up to multiplication of the profile by a constant,

$$\lambda_k = \left(\frac{k\pi}{L}\right)^2, \qquad B_k(x) = \sin\left(\frac{k\pi x}{L}\right), \qquad k \geq 1.$$

Solution : *If $\lambda = 0$, then B is affine and the two boundary conditions give $B = 0$. If $\lambda = -\mu^2 < 0$, then $B(x) = \alpha\cosh(\mu x) + \beta\sinh(\mu x)$. The condition at 0 gives $\alpha = 0$, and the one at L gives $\beta = 0$.*

For $\lambda = \mu^2 > 0$, one has $B(x) = \alpha\cos(\mu x) + \beta\sin(\mu x)$. Again $\alpha = 0$. A nonzero profile exists precisely when $\sin(\mu L) = 0$, that is, $\mu = k\pi/L$ for an integer $k \geq 1$. This gives the stated values and profiles.

3. **Temporal oscillations**
Define $\omega_k = ck\pi/L$. Solve the equation for the time-dependent factor and show that each normal mode has the form

$$u_k(t,x) = (a_k\cos(\omega_k t) + b_k\sin(\omega_k t))\sin\left(\frac{k\pi x}{L}\right). \tag{3}$$

Interpret ω_1 and the sequence $(\omega_k)_{k\geq 1}$.

Solution : *For $\lambda = \lambda_k$, the temporal equation is $A_k'' + \omega_k^2 A_k = 0$, hence $A_k(t) = a_k\cos(\omega_k t) + b_k\sin(\omega_k t)$. The fundamental angular frequency is $\omega_1 = c\pi/L$, and $\omega_k = k\omega_1$ are its harmonics. The corresponding frequencies in cycles per unit time are $f_k = \omega_k/(2\pi) = ck/(2L)$.*

4. **Finite sum solution**
Let $N \geq 1$ be a fixed integer. Suppose that the initial data are given by

$$u_0(x) = \sum_{k=1}^{N} \alpha_k \sin\left(\frac{k\pi x}{L}\right), \qquad v_0(x) = \sum_{k=1}^{N} \beta_k \sin\left(\frac{k\pi x}{L}\right).$$

By superposition, show that a solution of (2) can be written as

$$u(t,x) = \sum_{k=1}^{N} (a_k\cos(\omega_k t) + b_k\sin(\omega_k t))\sin\left(\frac{k\pi x}{L}\right). \tag{4}$$

Determine a_k and b_k in terms of α_k and β_k, and check the initial conditions.

Remark: For more general initial data, one would consider an infinite sum, leading to what is called a Fourier series. We only work with finite sums in this exercise.

Solution : *Each term in (4) satisfies the wave equation and the boundary conditions. Since the equation is linear, their finite sum does as well. At $t = 0$, we have*

$$u(0,x) = \sum_{k=1}^{N} a_k \sin\left(\frac{k\pi x}{L}\right), \qquad \partial_t u(0,x) = \sum_{k=1}^{N} \omega_k b_k \sin\left(\frac{k\pi x}{L}\right).$$

Thus we take $a_k = \alpha_k$ and $b_k = \beta_k/\omega_k$, which gives

$$u(t,x) = \sum_{k=1}^{N} \left(\alpha_k\cos(\omega_k t) + \frac{\beta_k}{\omega_k}\sin(\omega_k t)\right)\sin\left(\frac{k\pi x}{L}\right).$$

All derivatives are computed term by term, so this is a classical solution with exactly the prescribed initial data.

Exercise 3 — Equipartition of energy for the one-dimensional wave equation Let $c > 0$, $u_0 \in C_c^2(\mathbb{R})$ and $v_0 \in C_c^1(\mathbb{R})$, where the subscript c means that the functions have compact support. We consider

$$\begin{cases} \partial_{t,t} u - c^2 \partial_{x,x} u = 0, & t > 0, \quad x \in \mathbb{R}, \\ u(0,x) = u_0(x), \quad \partial_t u(0,x) = v_0(x), & x \in \mathbb{R}. \end{cases} \tag{5}$$

Assume that $\mathrm{supp}(u_0) \cup \mathrm{supp}(v_0) \subset [-R, R]$ for some $R > 0$. For $t \geq 0$, define the kinetic and potential energies by

$$K(t) = \frac{1}{2} \int_{\mathbb{R}} (\partial_t u(t,x))^2 \, \mathrm{d}x, \qquad P(t) = \frac{c^2}{2} \int_{\mathbb{R}} (\partial_x u(t,x))^2 \, \mathrm{d}x.$$

1. **Conservation of total energy**
   Prove that $E(t) = K(t) + P(t)$ is independent of time.

   Solution : *D'Alembert's formula gives $u(t,x) = 0$ when $|x| > R + ct$. On any bounded time interval $[0,T]$, the solution and the derivatives involved below therefore vanish outside a fixed compact interval. Since $u \in C^2$, differentiation under the integral sign and integration by parts are justified, with no boundary term. Thus*

   $$\begin{aligned} E'(t) &= \int_{\mathbb{R}} \partial_t u \, \partial_{t,t} u \, \mathrm{d}x + c^2 \int_{\mathbb{R}} \partial_x u \, \partial_{t,x} u \, \mathrm{d}x \\ &= \int_{\mathbb{R}} \partial_t u \left(\partial_{t,t} u - c^2 \partial_{x,x} u \right) \mathrm{d}x = 0. \end{aligned}$$

   Consequently $E(t) = E(0)$ for every $t \geq 0$.

2. **Traveling-wave profiles**
   Using d'Alembert's formula, write the solution as $u(t,x) = F(x+ct) + G(x-ct)$ and express F' and G' in terms of u_0' and v_0.

   Solution : *Let $V(x) = \int_0^x v_0(y) \, \mathrm{d}y$. D'Alembert's formula reads*

   $$u(t,x) = \frac{u_0(x+ct) + u_0(x-ct)}{2} + \frac{V(x+ct) - V(x-ct)}{2c}.$$

   We may therefore take $F = u_0/2 + V/(2c)$ and $G = u_0/2 - V/(2c)$, so that

   $$F' = \frac{1}{2} \left(u_0' + \frac{v_0}{c} \right), \qquad G' = \frac{1}{2} \left(u_0' - \frac{v_0}{c} \right).$$

   The derivatives F' and G' have support in $[-R,R]$, although F and G themselves need not have compact support.

3. **Separation and equipartition**
   Locate the supports of $F'(\cdot + ct)$ and $G'(\cdot - ct)$. Prove that, for every $t \geq R/c$,

   $$\int_{\mathbb{R}} (\partial_t u(t,x))^2 \, \mathrm{d}x = c^2 \int_{\mathbb{R}} (\partial_x u(t,x))^2 \, \mathrm{d}x. \tag{6}$$

   Deduce that $K(t) = P(t) = E(0)/2$ for $t \geq R/c$.

   Solution : *The support inclusions are*

   $$\mathrm{supp}\left(F'(\cdot + ct)\right) \subset [-R - ct, R - ct], \qquad \mathrm{supp}\left(G'(\cdot - ct)\right) \subset [-R + ct, R + ct].$$

   For $ct > R$, these intervals are disjoint. For $ct = R$, they meet only at $x = 0$, where both shifted derivatives vanish by continuity: a continuous function supported in $[-R,R]$ is zero at the endpoints. Thus the product $F'(x+ct)G'(x-ct)$ vanishes for every x when $ct \geq R$.

   Writing $f = F'(x+ct)$ and $g = G'(x-ct)$, we have $\partial_t u = c(f-g)$ and $\partial_x u = f + g$. Hence

   $$(\partial_t u)^2 - c^2 (\partial_x u)^2 = -4c^2 fg = 0 \qquad (t \geq R/c).$$

   Integration proves the identity. Therefore $K(t) = P(t)$, and conservation of E gives $K(t) = P(t) = E(0)/2$. The separation concerns the derivative profiles; the displacement u may still be nonzero between the two outgoing waves.

Exercise 4 — Numerical transport along Hamiltonian characteristics Consider a mass $m > 0$ attached to a spring of stiffness $k > 0$. If q denotes its displacement and $p = m\dot{q}$ its momentum, the equation $m\ddot{q} + kq = 0$ becomes

$$\dot{q} = \frac{p}{m}, \qquad \dot{p} = -kq, \qquad H(q,p) = \frac{1}{2}\left(\frac{p^2}{m} + kq^2\right).$$

This is the Hamiltonian system associated with H, as in Exercise 1. In the following, take $m = k = 1$, so that $H(q,p) = \frac{1}{2}(q^2 + p^2)$. The associated transport equation is

$$\partial_t u + p\,\partial_q u - q\,\partial_p u = 0, \qquad t > 0, \quad (q,p) \in \mathbb{R}^2. \tag{7}$$

Its characteristic curves satisfy $\dot{q} = p$ and $\dot{p} = -q$. For a time step $\tau > 0$, consider the following two approximations:

$$\text{Explicit Euler:} \quad \begin{cases} q_{n+1} = q_n + \tau p_n, \\ p_{n+1} = p_n - \tau q_n, \end{cases}$$

$$\text{Symplectic Euler:} \quad \begin{cases} p_{n+1} = p_n - \tau q_n, \\ q_{n+1} = q_n + \tau p_{n+1}. \end{cases}$$

1. **Preservation of area**
   Write each scheme as $X_{n+1} = MX_n$, where $X_n = (q_n, p_n)^\top$, and compute $\det M$. Which scheme preserves areas in the (q,p)-plane?

   Solution : *For explicit Euler, both updates use the old values, so*

   $$M_{\mathrm{E}} = \begin{pmatrix} 1 & \tau \\ -\tau & 1 \end{pmatrix}, \qquad \det M_{\mathrm{E}} = 1 + \tau^2.$$

   For symplectic Euler, substitution of the new momentum into the position update gives $q_{n+1} = (1 - \tau^2)q_n + \tau p_n$, *hence*

   $$M_{\mathrm{S}} = \begin{pmatrix} 1 - \tau^2 & \tau \\ -\tau & 1 \end{pmatrix}, \qquad \det M_{\mathrm{S}} = 1.$$

   A linear change of variables multiplies areas by the absolute value of its determinant. Thus explicit Euler expands areas by $1 + \tau^2$ *at each step, while symplectic Euler preserves them. More precisely, with* $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ *as in Exercise 1, direct multiplication gives* $M^\top J M = (\det M)J$ *for every real* 2×2 *matrix* M. *Consequently* $M_{\mathrm{S}}^\top J M_{\mathrm{S}} = J$, *which verifies the symplectic property in this setting.*

2. **Numerical comparison**
   Starting from $(q_0, p_0) = (1, 0)$, implement both schemes with $\tau = 0.1$ for 400 steps. Plot their trajectories in the (q,p)-plane and their energies $H(q_n, p_n)$ as functions of $t_n = n\tau$. Compare with the exact solution $(q(t), p(t)) = (\cos t, -\sin t)$.

   Solution : *Store the initial pair, then compute and store 400 successive pairs with each scheme. For explicit Euler, keep both old values until both new values have been computed. For symplectic Euler, compute the new momentum first and use it immediately in the position update. At every step, record* $H_n = (q_n^2 + p_n^2)/2$.

   The exact trajectory is the unit circle, traversed clockwise, and its energy is $1/2$. *For explicit Euler, expanding the two squares yields*

   $$H_{n+1} = (1 + \tau^2)H_n, \qquad H_n = \frac{1}{2}(1 + \tau^2)^n.$$

   Thus the numerical trajectory spirals outwards. At $n = 400$, $H_{400}/H_0 = 1.01^{400} \simeq 53.524$, *so* $H_{400} \simeq 26.762$.

   Symplectic Euler does not conserve H *exactly. However, it conserves the modified quadratic quantity*

   $$\widetilde{H}_\tau(q,p) = \frac{1}{2}(q^2 + p^2 - \tau qp).$$

Indeed, putting $p' = p - \tau q$ and $q' = q + \tau p'$, one obtains

$$\begin{aligned} q'^2 + p'^2 - \tau q' p' &= q^2 + p'^2 + \tau q p' \\ &= q^2 + p^2 - \tau q p. \end{aligned}$$

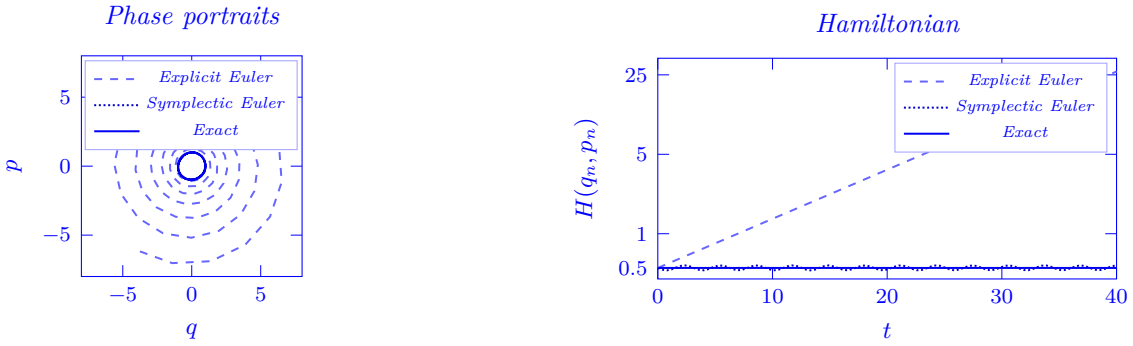
Since $2|qp| \le q^2 + p^2$, for $0 < \tau < 2$ we have

$$(1 - \tau/2)H(q,p) \le \widetilde{H}_\tau(q,p) \le (1 + \tau/2)H(q,p).$$

For the chosen initial pair, $\widetilde{H}_\tau(q_n, p_n) = 1/2$ for every n. With $\tau = 0.1$, this gives

$$\frac{1}{2(1.05)} \le H(q_n, p_n) \le \frac{1}{2(0.95)},$$

The lower and upper bounds are approximately 0.47619 and 0.52632. The numerical points stay on an ellipse close to the exact circle, and the physical energy has bounded oscillations. The numerical phase still differs from the exact phase over long times.



$(q_0, p_0) = (1, 0)$, $\tau = 0.1$, $N = 400$. *Explicit Euler spirals outwards. Symplectic Euler keeps H bounded but does not conserve it exactly; the right vertical axis is logarithmic.*

3. **Return to the transport equation**
Let $u_0 \in C^1(\mathbb{R}^2)$ be nonnegative and such that $\int_{\mathbb{R}^2} u_0(x)\,\mathrm{d}x < +\infty$. For each numerical matrix M, define

$$U^0 = u_0, \qquad U^{n+1}(x) = U^n(M^{-1}x).$$

Explain this approximation using characteristics. By a change of variables, show that

$$\int_{\mathbb{R}^2} U^{n+1}(x)\,\mathrm{d}x = |\det M| \int_{\mathbb{R}^2} U^n(x)\,\mathrm{d}x.$$

Which approximation preserves the total mass, as the exact transport equation does?

Solution : *The exact solution is constant along each characteristic. Replacing the exact advance over one time step by $x = My$ therefore gives $U^{n+1}(My) = U^n(y)$, which is the proposed update. Both matrices are invertible, and nonnegativity is preserved. With $x = My$, the change-of-variables formula from Exercise 1 gives*

$$\int_{\mathbb{R}^2} U^{n+1}(x)\,\mathrm{d}x = \int_{\mathbb{R}^2} U^{n+1}(My)|\det M|\,\mathrm{d}y = |\det M| \int_{\mathbb{R}^2} U^n(y)\,\mathrm{d}y.$$

This also proves finiteness of the mass at every step by induction. Writing $\mathcal{M}_n = \int_{\mathbb{R}^2} U^n(x)\,\mathrm{d}x$, we obtain

$$\mathcal{M}_n = (1 + \tau^2)^n \mathcal{M}_0 \quad \text{for explicit Euler,} \qquad \mathcal{M}_n = \mathcal{M}_0 \quad \text{for symplectic Euler.}$$

Thus symplectic Euler preserves the mass conservation law of the transport equation, while explicit Euler produces an artificial mass increase. The inverse matrices in this transport update are

$$M_{\mathrm{E}}^{-1} = \frac{1}{1+\tau^2}\begin{pmatrix} 1 & -\tau \\ \tau & 1 \end{pmatrix}, \qquad M_{\mathrm{S}}^{-1} = \begin{pmatrix} 1 & -\tau \\ \tau & 1 - \tau^2 \end{pmatrix}.$$

In particular, the inverse of the forward explicit Euler step is not obtained simply by replacing τ by $-\tau$ in that step.

4. **First-order convergence on a fixed time interval**
Assume that $u_0 \in C_c^1(\mathbb{R}^2)$ and fix $T > 0$. For either scheme, let U^n be defined as in Question 3, with $0 < \tau \leq 1$. Show that there exists a constant C_{T,u_0} such that

$$\max_{\substack{n\geq 0\\ n\tau\leq T}} \sup_{x\in\mathbb{R}^2} |U^n(x) - u(n\tau, x)| \leq C_{T,u_0}\tau. \tag{8}$$

The constant must be independent of τ and n. Both methods therefore converge with order one on every fixed time interval.

Hint. *Put* $B = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ *and use the matrix norm* $\|A\| = \sup_{|z|=1} |Az|$, *where* $|\cdot|$ *denotes the Euclidean norm. For either numerical matrix, one may use*

$$\|M\|,\ \|M^{-1}\| \leq 1 + 2\tau, \qquad \|M^{-1} - e^{-\tau B}\| \leq 2\tau^2,$$

as well as the identity

$$A^n - Q^n = \sum_{j=0}^{n-1} A^{n-1-j}(A - Q)Q^j.$$

First compare the inverse characteristic maps, then locate the supports of both solutions in a disk independent of τ *and* $n\tau \leq T$.

Solution : *Iterating Question 3 and using the exact characteristics gives*

$$U^n(x) = u_0(M^{-n}x), \qquad u(n\tau, x) = u_0(e^{-n\tau B}x).$$

The matrix e^{tB} *is a rotation, so its norm is* 1. *We first justify the estimates in the hint. For explicit Euler,* $M_{\mathrm{E}} = I_2 + \tau B$ *and* $M_{\mathrm{E}}^{-1} = (I_2 - \tau B)/(1 + \tau^2)$. *Since* $\|I_2 \pm \tau B\| = \sqrt{1 + \tau^2}$, *both norms are bounded by* $1 + 2\tau$. *Moreover,*

$$\|M_{\mathrm{E}}^{-1} - (I_2 - \tau B)\| = \frac{\tau^2}{\sqrt{1 + \tau^2}} \leq \tau^2.$$

For symplectic Euler, the matrices in Questions 1 and 3 give

$$M_{\mathrm{S}} = I_2 + \tau B - \tau^2 \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad M_{\mathrm{S}}^{-1} = I_2 - \tau B - \tau^2 \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Thus, for $0 < \tau \leq 1$, *both norms are at most* $\sqrt{1 + \tau^2} + \tau^2 \leq 1 + 2\tau$, *and* $\|M_{\mathrm{S}}^{-1} - (I_2 - \tau B)\| \leq \tau^2$. *Finally, the integral remainder in Taylor's formula yields*

$$e^{-\tau B} = I_2 - \tau B + \int_0^\tau (\tau - s)B^2 e^{-sB}\,\mathrm{d}s, \qquad \|e^{-\tau B} - (I_2 - \tau B)\| \leq \frac{\tau^2}{2}.$$

For either scheme, this proves the stated local error bound.

Apply the telescoping identity with $A = M^{-1}$ *and* $Q = e^{-\tau B}$. *Since* $\|Q^j\| = 1$, *for* $n \geq 1$ *and* $n\tau \leq T$ *we obtain*

$$\|M^{-n} - e^{-n\tau B}\| \leq 2\tau^2 \sum_{j=0}^{n-1} (1 + 2\tau)^{n-1-j} \leq 2n\tau^2 e^{2n\tau} \leq 2Te^{2T}\tau.$$

For $n = 0$, *the error is zero.*

Choose $R > 0$ *such that* u_0 *vanishes outside the disk* $|x| \leq R$. *The support of* U^n *is the image of the support of* u_0 *under the* forward *map* M^n. *Since* $\|M^n\| \leq (1 + 2\tau)^n \leq e^{2T}$, *this support lies in* $|x| \leq R_T$, *where* $R_T = Re^{2T}$. *The exact solution has its support in the disk* $|x| \leq R$, *because* $e^{n\tau B}$ *is a rotation. Hence both solutions vanish outside* $|x| \leq R_T$.

The first partial derivatives of u_0 *are bounded. Put*

$$L = \sup_{(q,p)\in\mathbb{R}^2} \sqrt{|\partial_q u_0(q,p)|^2 + |\partial_p u_0(q,p)|^2} < \infty.$$

Applying the fundamental theorem of calculus along a line segment gives $|u_0(y) - u_0(z)| \leq L|y - z|$ *for all* $y, z \in \mathbb{R}^2$. *For* $|x| \leq R_T$, *it follows that*

$$|U^n(x) - u(n\tau, x)| \leq L\|M^{-n} - e^{-n\tau B}\||x| \leq 2LR_T Te^{2T}\tau.$$

Outside this disk the difference is zero, which proves the uniform estimate with $C_{T,u_0} = 2LR_T Te^{2T}$.

Exercise 5 — Numerical schemes for the transport equation We compare two explicit finite-difference schemes for the transport equation. The aim is to observe numerical diffusion, identify an unstable scheme, and understand the role of the Courant number in these two phenomena. The computations may be performed with Python. For each plot, report the space step, the time step and the Courant number, and give a short interpretation of the result.

Let $L > 0$ and $a > 0$. We consider the periodic initial-value problem

$$\begin{cases} \partial_t u + a\partial_x u = 0, & t > 0, \quad 0 < x < L, \\ u(t,0) = u(t,L), & t \geq 0, \\ u(0,x) = u_0(x), & 0 \leq x < L. \end{cases} \tag{9}$$

Let $I \geq 2$ be an integer, $h = L/I$, $x_j = jh$ for $0 \leq j < I$, and $t_n = n\Delta t$. We write U_j^n for an approximation of $u(t_n, x_j)$ and define the Courant number $\nu = a\Delta t/h$. Spatial indices are understood modulo I. The upwind scheme is

$$U_j^{n+1} = (1-\nu)U_j^n + \nu U_{j-1}^n, \tag{10}$$

whereas the centered explicit scheme is

$$U_j^{n+1} = U_j^n - \frac{\nu}{2}\left(U_{j+1}^n - U_{j-1}^n\right). \tag{11}$$

For the experiments, take $L = a = 1$ and start with $I = 200$.

1. **Numerical diffusion in the upwind scheme**
   Implement (10) with $U_j^0 = u_0(x_j)$. Use separately the smooth periodic profile

$$u_0(x) = \exp\left(-\frac{200}{\pi^2}\sin^2(\pi(x-0.3))\right)$$

   and the periodic extension of the discontinuous profile

$$u_0(x) = \begin{cases} 1, & 0.2 \leq x \leq 0.4, \\ 0, & \text{otherwise on } [0,1). \end{cases}$$

   For each profile, compare the numerical solution with the exact translated solution after one period, at $T = L/a$, for $\nu \in \{0.25, 0.5, 0.9, 1\}$. If T is not a grid time, shorten only the last time step to reach T, replacing ν on that step by $\nu_{\text{last}} = a(T - t_N)/h$, where $t_N < T < t_{N+1}$. Report its value. Describe the numerical diffusion and its dependence on ν.

   Solution : *Store the I values in an array and form a second array from the old values using (10); updating entries one by one in the same array would change the scheme. The value preceding U_0^n is U_{I-1}^n. At each step use $\delta t = \min(\Delta t, T - t)$ and $\nu_{\text{step}} = a\delta t/h$. For $I = 200$, the nominal time steps are 0.00125, 0.0025, 0.0045 and 0.005. Only $\nu = 0.9$ requires a shorter last step: after 222 full steps, $t_{222} = 0.999$, so $\delta t = 0.001$ and $\nu_{\text{last}} = 0.2$.*

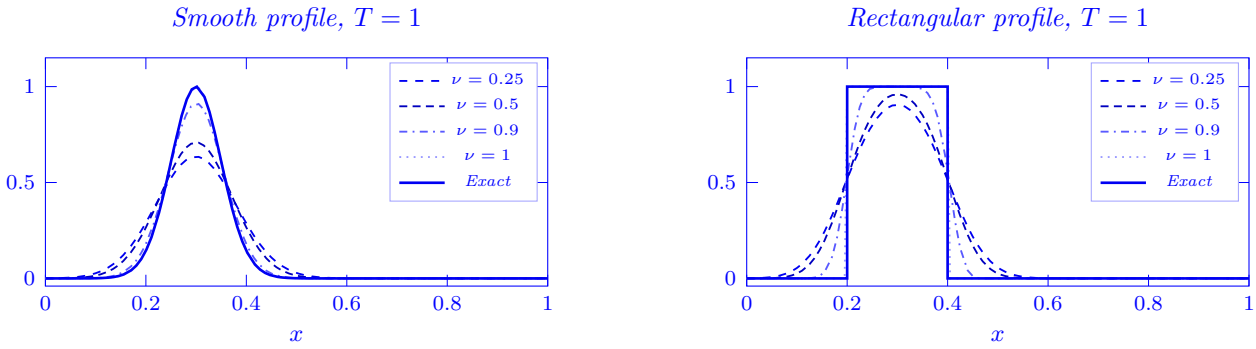
   The exact solution is $u(t,x) = u_0((x - at) \bmod L)$, hence $u(T,x) = u_0(x)$. When $0 \leq \nu \leq 1$, the update is an average of two neighboring values. It preserves $h\sum_j U_j^n$ and gives $\max_j |U_j^{n+1}| \leq \max_j |U_j^n|$. To quantify the smoothing, consider a discrete Fourier mode $U_j^n = G^n \exp(\mathrm{i}j\theta)$, where $\theta = 2\pi\ell/I$ and $0 \leq \ell < I$. Substitution gives

$$G(\theta) = 1 - \nu + \nu\exp(-\mathrm{i}\theta), \qquad |G(\theta)|^2 = 1 - 4\nu(1-\nu)\sin^2(\theta/2).$$

   Nonconstant modes are damped when $0 < \nu < 1$. Over a fixed physical time, the damping becomes weaker as ν approaches 1. More precisely, after n equal steps the scheme averages translated values with binomial weights. Before reducing positions modulo L, the variance of these displacements is $h^2 n\nu(1-\nu) = aht_n(1-\nu)$. Thus the smooth peak broadens and decreases, while the two edges of the rectangular profile become rounded. For a smooth profile, a formal *Taylor expansion as $h \to 0$ with ν fixed gives the modified equation*

$$u_t + au_x = \frac{ah}{2}(1-\nu)u_{xx} + O(h^2).$$

   This is a formal explanation of the diffusion coefficient. The Taylor calculation requires smoothness and does not apply to the discontinuous profile, whose smoothing is instead explained by the averaging in the scheme.



Periodic transport, $I = 200$. The last step is shortened when needed. For $\nu = 1$, the final grid values equal the initial values.

2. **Exact translation when $\nu = 1$**
   Prove directly from (10) that, when $\nu = 1$, $U_j^n = U_{j-n}^0$. Check this identity numerically for both initial profiles, and verify that the grid values return to their initial values after one period, up to round-off errors.

   Solution : *For $\nu = 1$, the update is $U_j^{n+1} = U_{j-1}^n$. Induction gives $U_j^n = U_{j-n}^0$, with indices modulo I. In particular, after I steps, $t_I = Ih/a = L/a$ and $U_j^I = U_j^0$. A useful check is $\max_j |U_j^I - U_j^0|$; copying array entries gives zero exactly in this case. The numerical solution agrees with the exact solution at every grid point and grid time, for either initial profile.*

3. **Instability of the centered scheme**
   Implement (11). Add a random perturbation bounded by 10^{-8} to the smooth initial profile. Compare the upwind and centered schemes for $\nu = 0.8$ over $0 \le t \le T = 1$, using the same perturbed data. Plot the solution profiles at several times and $\max_j |U_j^n|$ as a function of time. Describe the oscillations and determine whether the condition $a\Delta t \le h$ prevents their growth.

   Solution : *Choose independent numbers $\eta_j \in [-1, 1]$ and initialize $U_j^0 = u_0(x_j) + 10^{-8}\eta_j$. Form both neighboring arrays from the old values before applying (11). Here $\Delta t = 0.004$; 250 steps reach $T = 1$. For the centered scheme the amplification factor is*

$$G(\theta) = 1 - \mathrm{i}\nu \sin\theta, \qquad |G(\theta)|^2 = 1 + \nu^2 \sin^2\theta.$$

Hence every mode with $\sin\theta \neq 0$ is amplified for $\nu \neq 0$. The greatest amplification occurs near $\theta = \pi/2$; for $\nu = 0.8$ it is $\sqrt{1.64} \simeq 1.281$ per step. The alternating mode $\theta = \pi$ is neutral. The perturbation excites the growing modes, which eventually dominate the transported profile and make the maximum grow rapidly. A logarithmic vertical scale makes this growth easier to read. In contrast, the upwind maximum stays bounded by its initial value, although the peak is smoothed. Satisfying $a\Delta t \le h$ therefore does not suffice to stabilize the centered explicit transport scheme.

Exercise 6 — Numerical experiments for the wave equation We study the propagation of two pulses, compare reflection with absorption at the endpoints, and observe the instability caused by violating the CFL condition. The computations may be performed with Python. For each plot, report the space step, time step and Courant number, and give a short interpretation of the result. We first consider a string with fixed endpoints:

$$\partial_{t,t} u - c^2 \partial_{x,x} u = 0, \qquad t > 0, \quad 0 < x < L, \qquad u(t,0) = u(t,L) = 0. \tag{12}$$

Let $I \ge 2$ be an integer, $h = L/I$, $x_j = jh$ for $0 \le j \le I$ and $t_n = n\Delta t$. We write U_j^n for an approximation of $u(t_n, x_j)$ and define the Courant number $\lambda = c\Delta t/h$. The centered scheme is

$$U_j^{n+1} = 2(1-\lambda^2)U_j^n + \lambda^2\left(U_{j-1}^n + U_{j+1}^n\right) - U_j^{n-1}, \qquad 1 \le j \le I-1, \tag{13}$$

with $U_0^n = U_I^n = 0$. We take $u(0,x) = u_0(x)$ and $\partial_t u(0,x) = 0$. Set $U_j^0 = u_0(x_j)$ and initialize the first step by

$$U_j^1 = U_j^0 + \frac{\lambda^2}{2}\left(U_{j+1}^0 - 2U_j^0 + U_{j-1}^0\right), \qquad 1 \le j \le I-1. \tag{14}$$

Take $L = c = 1$, initially $I = 400$, and

$$u_0(x) = \begin{cases} \exp\left(-\dfrac{1}{1-((x-1/2)/r)^2}\right), & |x-1/2| < r, \\ 0, & |x-1/2| \ge r, \end{cases} \qquad r = 0.15. \tag{15}$$

1. **Propagation, reflection and energy**
Implement (13) with $\lambda = 0.9$. Observe the splitting of the initial displacement into two traveling waves and their reflections at the endpoints, over $0 \le t \le 2L/c$. Plot the numerical energy

$$E^n = \frac{h}{2}\sum_{j=1}^{I-1}\left(\frac{U_j^{n+1}-U_j^n}{\Delta t}\right)^2 + \frac{c^2 h}{2}\sum_{j=0}^{I-1}\left(\frac{U_{j+1}^n - U_j^n}{h}\right)^2.$$

Is it approximately conserved? Refine the mesh, keeping λ fixed, and compare the variations of E^n over the same physical time interval.

Solution : *Initialize the array U^0 from (15), then compute U^1 using (14). Store two successive arrays and compute each new array using (13), always setting its first and last entries to zero. Compute E^n before discarding U^n. The initial profile must be evaluated only inside its support, to avoid division by zero at the edges. Here $h = 0.0025$ and $\Delta t = 0.00225$; run until the first grid time at least 2.*

Before the waves reach the boundary, d'Alembert's formula gives $u(t,x) = \frac{1}{2}u_0(x-ct) + \frac{1}{2}u_0(x+ct)$. The two pulses therefore travel at speed c with half the initial amplitude. They begin reaching the endpoints at $t = 0.35$. Reflection at a fixed endpoint reverses their sign. Using the odd, $2L$-periodic extension of u_0 in d'Alembert's formula gives the exact solution after reflection; for this symmetric profile, $u(L/c, x) = -u_0(x)$ and $u(2L/c, x) = u_0(x)$.

The displayed E^n is only approximately conserved. An exactly conserved quantity for the scheme is instead

$$\widetilde{E}^{n+1/2} = \frac{h}{2}\sum_{j=1}^{I-1}\left(v_j^{n+1/2}\right)^2 + \frac{c^2 h}{2}\sum_{j=0}^{I-1} D_h U_j^{n+1} D_h U_j^n,$$

where $v_j^{n+1/2} = (U_j^{n+1} - U_j^n)/\Delta t$ and $D_h U_j^n = (U_{j+1}^n - U_j^n)/h$. To verify this, multiply the scheme, written as

$$\frac{U_j^{n+1} - 2U_j^n + U_j^{n-1}}{\Delta t^2} = c^2 \frac{U_{j+1}^n - 2U_j^n + U_{j-1}^n}{h^2},$$

by $h(U_j^{n+1} - U_j^{n-1})/(2\Delta t)$ and sum over j. The left-hand side is the difference of the two kinetic energies divided by Δt. On the right, the discrete integration by parts identity

$$h\sum_{j=1}^{I-1}\frac{A_{j+1} - 2A_j + A_{j-1}}{h^2}B_j = -h\sum_{j=0}^{I-1} D_h A_j D_h B_j$$

applies when $A_0 = A_I = B_0 = B_I = 0$. It gives $\widetilde{E}^{n+1/2} = \widetilde{E}^{n-1/2}$.

This energy also explains stability. Put $M^{n+1/2} = (U^{n+1} + U^n)/2$ and write $\|A\|_h^2 = h\sum_j A_j^2$, summing over the corresponding grid. Then

$$\widetilde{E}^{n+1/2} = \frac{1}{2}\|v^{n+1/2}\|_h^2 - \frac{c^2\Delta t^2}{8}\|D_h v^{n+1/2}\|_h^2 + \frac{c^2}{2}\|D_h M^{n+1/2}\|_h^2.$$

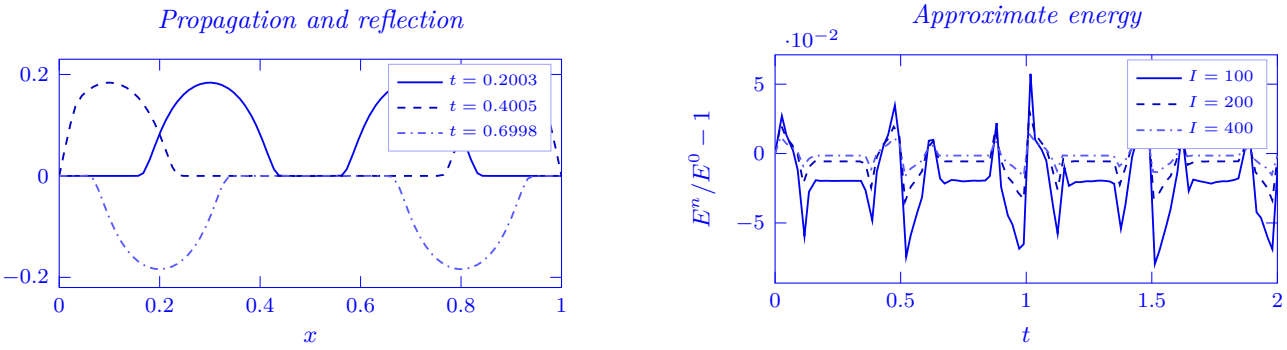
Since $\|D_h v\|_h^2 \le 4h^{-2}\|v\|_h^2$, it follows that

$$\widetilde{E}^{n+1/2} \ge \frac{1-\lambda^2}{2}\|v^{n+1/2}\|_h^2 + \frac{c^2}{2}\|D_h M^{n+1/2}\|_h^2.$$

This gives a positive energy for $\lambda < 1$. Finally,

$$E^n - \widetilde{E}^{n+1/2} = -\frac{c^2 h\Delta t}{2}\sum_{j=0}^{I-1} D_h U_j^n D_h v_j^{n+1/2}.$$

For a smoothly resolved solution on a fixed time interval, this difference is small, of order Δt when the two discrete gradients remain uniformly bounded. Thus E^n has small oscillations that decrease under refinement, whereas $\widetilde{E}^{n+1/2}$ stays constant up to round-off errors. Compare, for instance, $I = 100, 200, 400$ using $\Delta t = 0.9h/c$ in each case.



Waves, $\lambda = 0.9$. Reflected pulses change sign. The variations of E^n decrease with h; the cross-time energy $\widetilde{E}^{n+1/2}$ is constant up to round-off errors.

2. **CFL condition and instability**
Return to fixed endpoints, $U_0^n = U_I^n = 0$, and repeat the simulation with $\lambda = 1.05$. Add a random perturbation bounded by 10^{-8} to the *interior* entries of U^0, keeping $U_0^0 = U_I^0 = 0$, and then initialize U^1 by (14). Compare with $\lambda = 0.9$ using the same perturbed data, and plot $\max_j |U_j^n|$. Relate the observations to the CFL condition $c\Delta t \le h$.

Solution : *Clamp both endpoints to zero after every update. At $I = 400$, $\lambda = 1.05$ gives $\Delta t = 0.002625$. Use a logarithmic vertical scale for the maximum and stop the unstable computation before floating-point overflow. The stable solution propagates and reflects, whereas the unstable solution soon develops rapidly growing oscillations.*

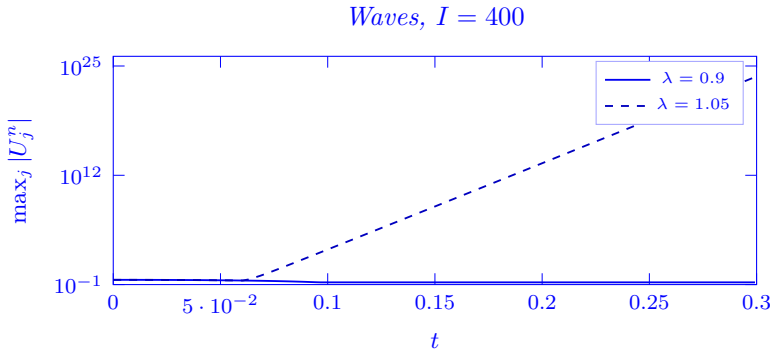
To justify the instability, consider a discrete sine mode $U_j^n = A^n \sin(k\pi j/I)$, $1 \le k \le I-1$. Its amplitude satisfies

$$A^{n+1} - 2(1 - 2\sigma_k^2)A^n + A^{n-1} = 0, \qquad \sigma_k = \lambda \sin\left(\frac{k\pi}{2I}\right).$$

If $\sigma_k < 1$, the two characteristic roots have modulus one. If $\sigma_k > 1$, one root has modulus

$$2\sigma_k^2 - 1 + 2\sigma_k\sqrt{\sigma_k^2 - 1} > 1.$$

On this grid, for $k = I-1$, $\sigma_{I-1} = 1.05\cos(\pi/800) \simeq 1.04999 > 1$; the growing root has modulus approximately 1.877. The perturbation makes this unstable component visible. Thus the grid and parameters used here do produce instability. For a general fixed grid, $\lambda > 1$ produces a growing mode precisely when $\lambda\cos(\pi/(2I)) > 1$; every fixed $\lambda > 1$ does so on sufficiently fine grids. The CFL condition $c\Delta t \le h$ prevents this exponential modal growth on every grid.



Initial perturbation of amplitude 10^{-8}; logarithmic vertical scale. The unstable wave scheme rapidly amplifies oscillations.

3. **Absorbing boundary conditions**
Replace the fixed endpoint conditions by

$$\partial_t u(t,0) - c\partial_x u(t,0) = 0, \qquad \partial_t u(t,L) + c\partial_x u(t,L) = 0. \tag{16}$$

Check that a wave traveling to the left satisfies the first condition, and a wave traveling to the right satisfies the second one. Keep the same initial displacement, zero initial velocity and $\lambda = 0.9$. Initialize

U^0 and the interior of U^1 as in Question 1, with $U_0^1 = U_I^1 = 0$ since the initial bump vanishes near the endpoints. For $n \geq 1$, compute the new interior entries using (13), and then set

$$\begin{aligned} U_0^{n+1} &= U_1^n + \rho(U_1^{n+1} - U_0^n), \\ U_I^{n+1} &= U_{I-1}^n + \rho(U_{I-1}^{n+1} - U_I^n), \qquad \rho = \frac{\lambda - 1}{\lambda + 1}. \end{aligned} \tag{17}$$

Explain how these formulas approximate the two boundary conditions. Compare solution profiles and $\max_j |U_j^n|$ with the fixed endpoint simulation, over $0 \leq t \leq 2L/c$. When do the waves leave the interval? Does the numerical solution disappear exactly? Refine the mesh to test the remaining amplitude.

Solution : *A left-going wave has the form $F(x + ct)$, so that $u_t - cu_x = 0$. A right-going wave has the form $G(x - ct)$, so that $u_t + cu_x = 0$. The signs in (16) therefore let the outgoing waves leave, without creating a reflected wave in the continuous equation.*

To obtain the left boundary update, approximate both derivatives at the middle of the first space-time cell:

$$\frac{U_0^{n+1} + U_1^{n+1} - U_0^n - U_1^n}{2\Delta t} - c\frac{U_1^{n+1} + U_1^n - U_0^{n+1} - U_0^n}{2h} = 0.$$

Solving for U_0^{n+1} gives (17). The right boundary formula follows in the same way with the opposite sign of u_x. Thus first compute U_j^{n+1} for $1 \leq j \leq I - 1$, and only then compute the two endpoint entries. Do not also set them to zero: that would restore the fixed endpoint conditions. Initially, $U_0^0 = U_I^0 = 0$. For the compact bump and grids used here, U^0 vanishes near both endpoints, so the zero-velocity first step has $U_0^1 = U_I^1 = 0$ as well; thereafter use the outgoing updates.

Extend the initial bump by zero outside $[0, L]$. The exact solution with these boundary conditions is

$$u(t, x) = \frac{1}{2}u_0(x - ct) + \frac{1}{2}u_0(x + ct), \qquad 0 \leq x \leq L.$$

Each pulse leaves as soon as it reaches an endpoint. For $L = c = 1$ and $r = 0.15$, the exits begin at $t = 0.35$ and are complete at $t = 0.65$. The continuous solution is therefore zero for $t \geq 0.65$. In contrast, fixed endpoints reflect both pulses with reversed sign. The corresponding continuous energy satisfies

$$\frac{\mathrm{d}}{\mathrm{d}t}\left[\frac{1}{2}\int_0^L (u_t^2 + c^2u_x^2)\,\mathrm{d}x\right] = c^2[u_t u_x]_0^L = -c(u_t(t,0)^2 + u_t(t,L)^2) \leq 0.$$

The loss of energy is the energy carried out of the interval. The fixed endpoint discrete energy identity in Question 1 must not be applied after changing the boundary conditions.

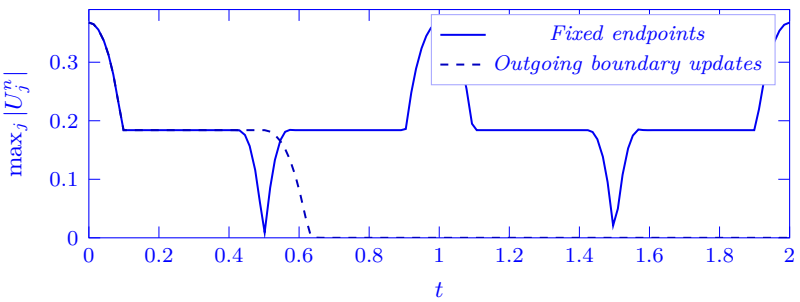
The numerical absorbing conditions are approximate. They do not amplify propagating discrete waves when $0 < \lambda < 1$, but a small reflected part generally remains. To see the distinction, a mode of spatial phase $0 < \theta < \pi$ has temporal phase $\omega\Delta t$ given by $\sin(\omega\Delta t/2) = \lambda \sin(\theta/2)$. Substitution of an outgoing mode and its reflected mode into (17) gives the reflection amplitude

$$|R(\theta)| = \frac{\sqrt{1 - \lambda^2 \sin^2(\theta/2)} - \cos(\theta/2)}{\sqrt{1 - \lambda^2 \sin^2(\theta/2)} + \cos(\theta/2)} \leq 1.$$

It is small for well-resolved waves, but generally nonzero for $\lambda = 0.9$. At $\lambda = 1$ the endpoint formulas reduce to exact one-cell outgoing shifts. Thus the plots should show almost complete departure, with small dispersive tails rather than exact disappearance on the grid. For example, the maximum remaining amplitude over $0.8 \leq t \leq 2$ is approximately

I	200	400	800
$\max_{0.8 \leq t_n \leq 2} \max_j \lvert U_j^n \rvert$	8.6×10^{-5}	2.6×10^{-5}	6.9×10^{-6}.

These values are numerical checks, with $\Delta t = 0.9/I$; the initial maximum is approximately 0.368.



$I = 400$, $\lambda = 0.9$. *The reflected pulses remain in the string with fixed endpoints; outgoing boundary updates let the pulses leave, with a small numerical residual.*